STRANGER SYNTAX

Mind-Flayer Patch v2

⚠️ CONFIDENTIAL BRIEFING

Clearance Level: Hawkins Lab – Level 3

A large-scale digital platform is facing an escalating crisis.

Content circulating on the platform is causing real-world harm — not always through false information, but through manipulation of context, framing, and narrative.
The system is under active observation by adversarial actors who adapt to detection strategies in real time.

STRANGER SYNTAX

You have been recruited to design a resilient detection and mitigation system that can operate under technical, ethical, financial, and human constraints.

There is no single correct answer.
You will be evaluated on reasoning, trade-offs, and system resilience.

🎯 YOUR MISSION

Design an end-to-end system that can:

1. Detect potentially harmful or misleading content
2. Decide when to act automatically, when to escalate, and when to wait
3. Balance speed, accuracy, cost, and human trust
4. Remain effective as adversaries adapt
5. Your solution must be defensible, cost-aware, and adaptive.

🧠 SYSTEM CONSTRAINTS

(All constraints are mandatory and enforced)

1️⃣ Signal Collapse Rule (Adaptive Adversary)

Every detection method becomes less effective if overused.
If a detection technique is triggered too frequently, adversaries learn to bypass it.

Effect:
Each detection method has a decay factor applied to its confidence when overused.

> Over-reliance on a single model or technique will degrade system performance over time.

2️⃣ Partial Observability (Latency Trade-off)

STRANGER SYNTAX

Content does not arrive all at once.

Information is revealed in stages:

- **Stage 1**: Metadata only
- **Stage 2:** Partial content (text OR thumbnail, not both)
- **Stage 3:** Full content (only if escalated)

⚠️ Waiting for full content too often violates latency requirements and incurs penalties.

3️⃣ Human Trust Budget (Hard Limit)

- You have access to only 20 human reviews per hour.
- Human review improves accuracy
- Overuse causes delays and backlogs
- Incorrect automated actions reduce platform trust
- You must decide what truly deserves human attention.

4️⃣ Multilingual Ambiguity (Context Trap)

The platform includes Urdu, Roman Urdu, and English content.

Some phrases are context-dependent and cannot be safely classified without surrounding information.

Example:

> “yeh fire hai”
(praise OR threat depending on context)

⚠️ Hard classification of ambiguous content is penalized.

5️⃣ Poisoned Feedback Loop

- Human labels are not always reliable.
- Approximately 15% of human moderation labels contain noise or bias
- Some feedback may be coordinated or adversarial
- Blindly trusting labels can reinforce harmful patterns.

STRANGER SYNTAX

6️⃣ Budget Reality (No Infinite Compute)

You are given a monthly system budget of **PKR 50,000**.

This must cover:

- Model inference
- API usage
- Any automated analysis

If the budget is exceeded, parts of the system must shut down.

📦 REQUIRED DELIVERABLES

Your team must prepare the following:

1️⃣ System Architecture

- High-level diagram
- Major components and data flow

2️⃣ Decision Logic

- How content moves through the system
- When actions are taken vs deferred

3️⃣ Cost & Latency Strategy

STRANGER SYNTAX

- Estimated monthly cost
- How latency constraints are respected

4️⃣ Adaptation Strategy

- How your system responds when detection methods degrade
- How it avoids becoming predictable

5️⃣ Failure Mode Declaration (Mandatory)

Clearly state:

- Three scenarios where your system will fail
- The potential impact of each failure
- How damage is detected or contained

🏆 EVALUATION CRITERIA

Your solution will be assessed on:

1. Soundness of reasoning and trade-offs
2. Adaptability to adversarial behavior
3. Awareness of cost and latency
4. Handling of ambiguity and ethics
5. Clarity and confidence in explanation

⚠️ Over-engineered or buzzword-heavy solutions without justification will be challenged.

STRANGER SYNTAX

🧩 IMPORTANT NOTES

- There is no perfect solution
- Simplicity with strong reasoning is valued
- You may question assumptions, but must defend alternatives
- This is a thinking challenge, not a coding contest

🕯️ FINAL MESSAGE

“The enemy learns.
Static systems fail.
Only adaptive thinking survives.”

Friends don’t lie. Good luck.

STRANGER SYNTAX